

Week 2: Exploratory Data Analysis - NextGenLearners Applicant Insights

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Dataset: applicants_cleaned.csv

This notebook analyzes applicant data to understand domain popularity, application trends, acceptance rates, and university distribution to provide actionable insights for the NextGenLearners program.

```
In [27]: # import libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [28]: # Load the dataset
df = pd.read_csv('applicants_cleaned.csv', parse_dates = ['Application Date'])
```

```
In [29]: # # data types + non-null counts per column
df.info()
# preview the first 5 rows
df.head()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 72 entries, 0 to 71
Data columns (total 7 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Applicant Name        72 non-null    str
1   Email                 72 non-null    str
2   Phone                 72 non-null    str
3   Domain Applied        72 non-null    str
4   University            72 non-null    str
5   Application Date      66 non-null    datetime64[us]
6   Status                72 non-null    str
dtypes: datetime64[us](1), str(6)
memory usage: 4.1 KB
```

Out[29]:

	Applicant Name	Email	Phone	Domain Applied	University	Age
0	Adam Miller	bcherry@example.com	+12794198633x555	Web Development	Saunders Inc University	20
1	Amanda White	gmcgrath@example.org	9877971502	UI/UX Design	Oneal Inc University	20
2	Emily Torres	villanuevathomas@example.net	(620)3613890x4883	Data Science	Petty, Clark And Irwin University	20
3	James Robinson	williamvalentine@example.org	4812129695	Web Development	Ho, Garcia And Gutierrez University	20
4	Tina Newton	kharris@example.net	(795)2683086x8476	AI/ML	Williams, Murphy And Lewis University	20

Graph 1: Which Domain Received the Most Applications?

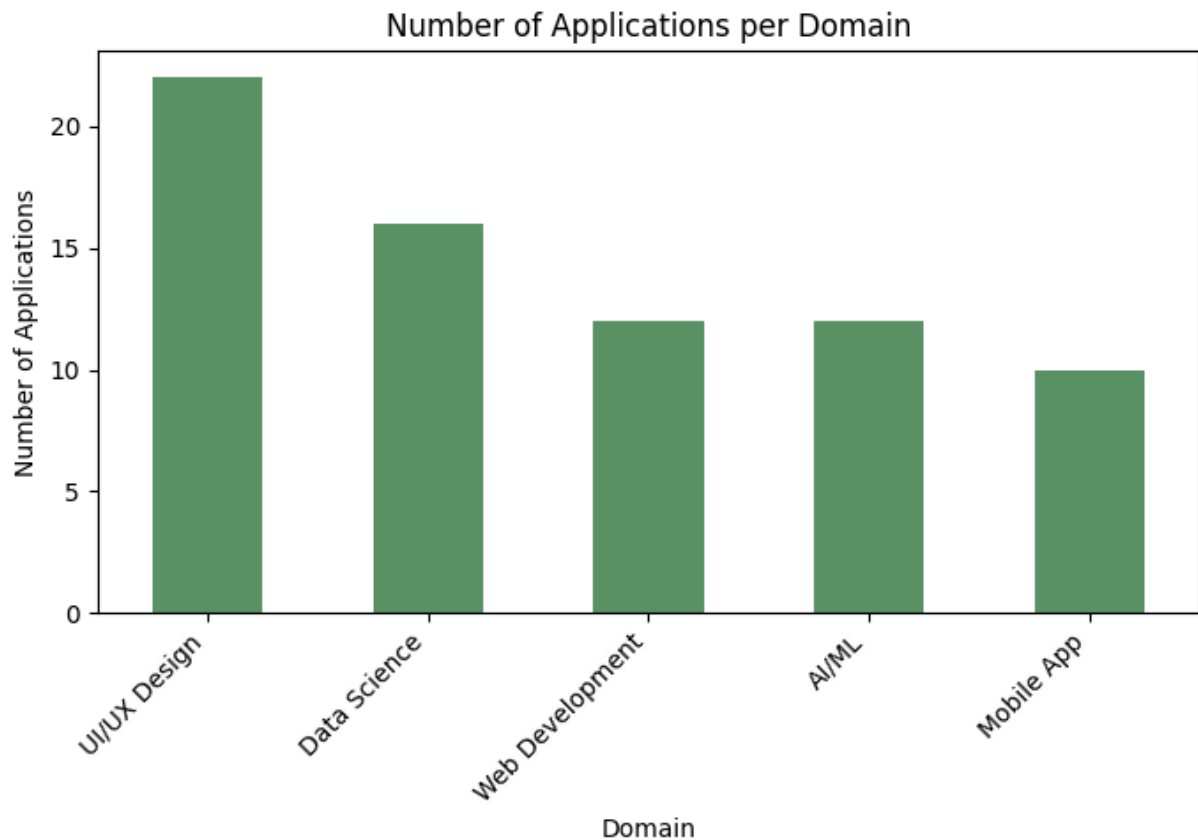
In [30]:

```
domain_counts = df['Domain Applied'].value_counts()
print(domain_counts)
```

Domain Applied
UI/UX Design 22
Data Science 16
Web Development 12
AI/ML 12
Mobile App 10
Name: count, dtype: int64

In [31]:

```
plt.figure(figsize=(7,5))
domain_counts.plot(kind='bar', color='#5A9364')
plt.title('Number of Applications per Domain')
plt.xlabel('Domain')
plt.ylabel('Number of Applications')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()
```



Takeaway: UI/UX Design received the highest number of applications (22), while Mobile App received the lowest (10).

plt.title('Number of Applications per Domain'): This sets the main title of the chart. It tells the reader what the graph is about.

plt.xlabel('Domain'): This labels the X-axis. It shows what categories are on the horizontal axis. Here it's

plt.xticks(rotation=45, ha='right'): This rotates the X-axis labels by 45 degrees to the right. We do this so long labels like "Web Development" don't overlap and become unreadable.

Graph 2: How Did Applications Trend Over Time?

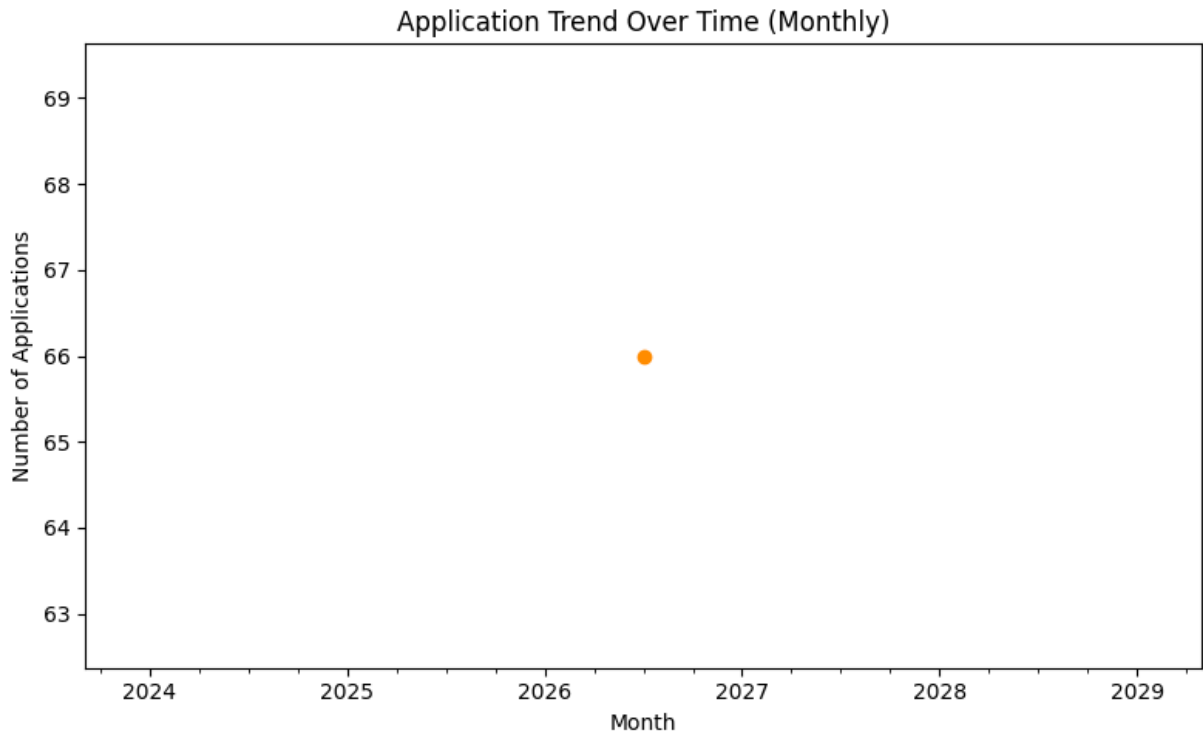
```
In [32]: # Group by month
df['Application Month'] = df['Application Date'].dt.to_period('M')
monthly_trend = df.groupby('Application Month').size()
print(monthly_trend)
```

```
Application Month
2026-07      66
Freq: M, dtype: int64
```

```
In [33]: plt.figure(figsize=(8,5))
monthly_trend.plot(kind='line', marker='o', color='darkorange')
```

```
plt.title('Application Trend Over Time (Monthly)')
plt.xlabel('Month')
plt.ylabel('Number of Applications')
plt.tight_layout()
plt.show()
```

C:\Users\lenovo\AppData\Local\Programs\Python\Python313\Lib\site-packages\pandas\plotting_matplotlib\core.py:1613: UserWarning: Attempting to set identical low and high xlims makes transformation singular; automatically expanding.
 ax.set_xlim(left, right)



Observation: All 66 applications were submitted in July 2026.

Takeaway: The dataset covers only 1 month, so no trend can be observed yet. More data is needed.

Graph 3: What's the Acceptance Rate - Overall and Per Domain?

Overall status breakdown:

```
In [34]: status_counts = df['Status'].value_counts()
print(status_counts)
```

```
Status
Rejected      29
Under Review   27
Selected       16
Name: count, dtype: int64
```

Per-domain breakdown using pivot_table():

```
In [35]: status_by_domain = pd.pivot_table(df, index='Domain Applied', columns='Status', values='Count',
print(status_by_domain)
```

Status	Rejected	Selected	Under Review
Domain Applied			
AI/ML	4	5	3
Data Science	9	2	5
Mobile App	3	3	4
UI/UX Design	6	6	10
Web Development	7	0	5

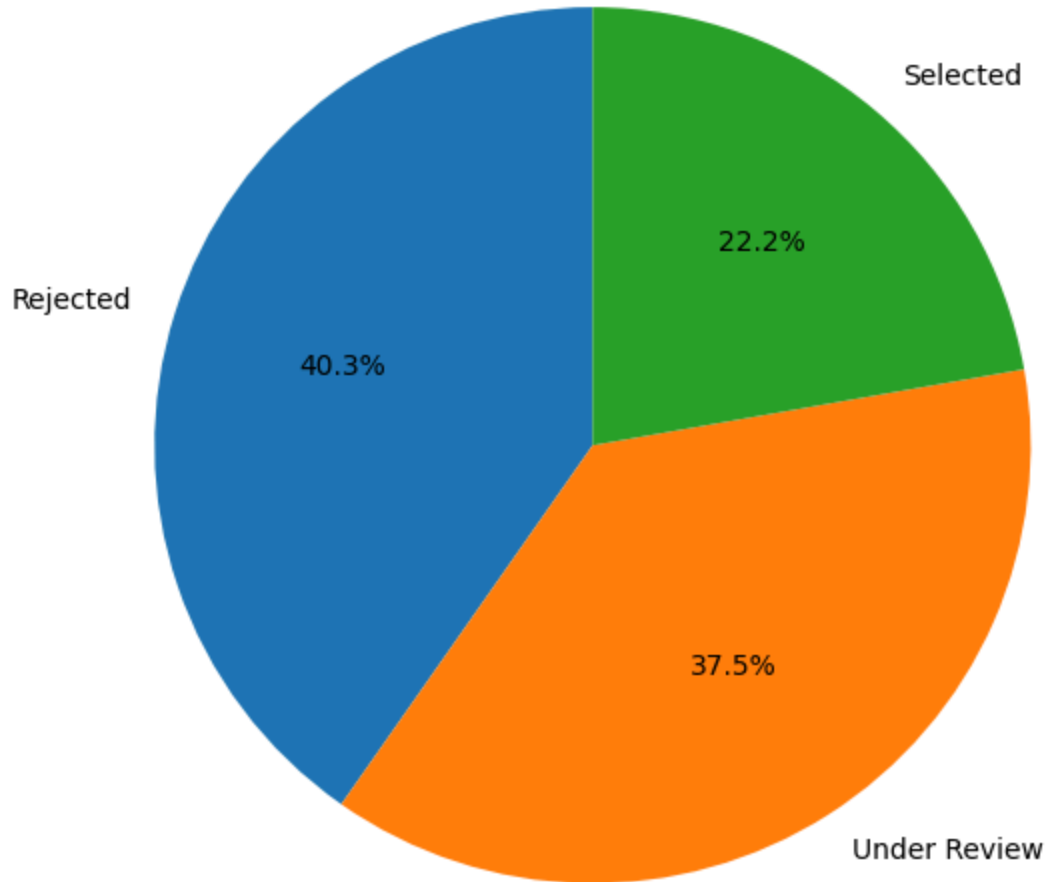
Calculate acceptance rate (percentage), not just raw counts:

```
In [36]: status_by_domain_pct = status_by_domain.div(status_by_domain.sum(axis=1),
axis=0) * 100
print(status_by_domain_pct.round(1))
```

Status	Rejected	Selected	Under Review
Domain Applied			
AI/ML	33.3	41.7	25.0
Data Science	56.2	12.5	31.2
Mobile App	30.0	30.0	40.0
UI/UX Design	27.3	27.3	45.5
Web Development	58.3	0.0	41.7

```
In [37]: # Overall status - pie chart
plt.figure(figsize=(6,6))
status_counts.plot(kind='pie', autopct='%1.1f%%', startangle=90)
plt.title('Overall Application Status Breakdown')
plt.ylabel('') # removes the default awkward y-label pie charts add
plt.tight_layout()
plt.show()
```

Overall Application Status Breakdown

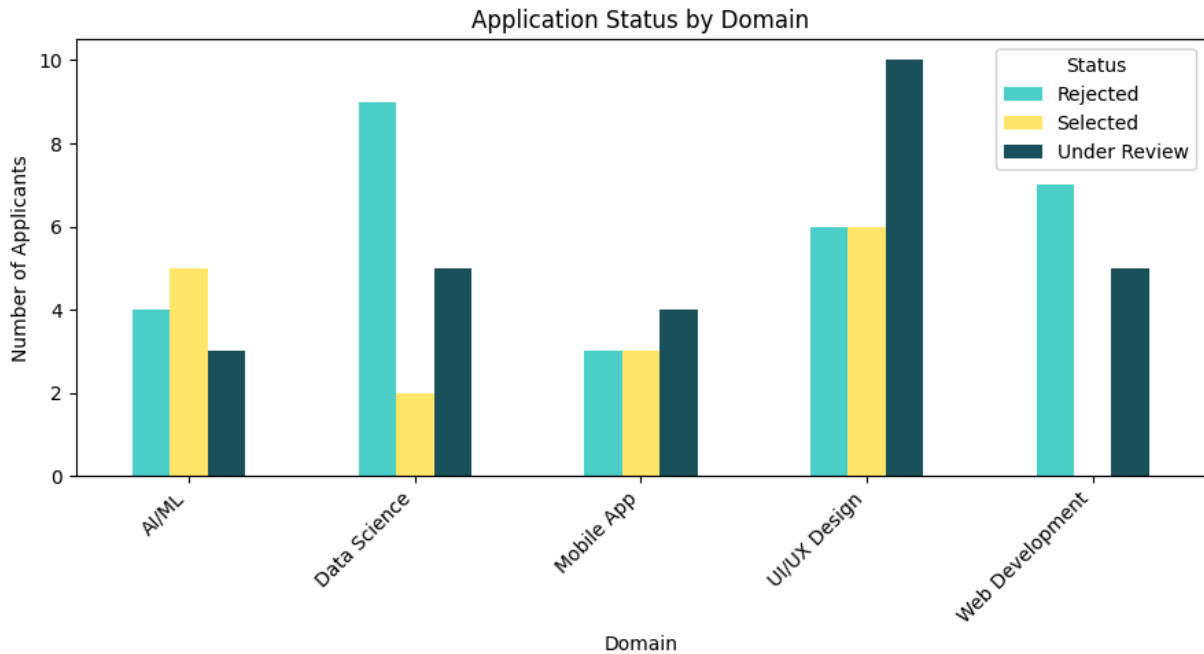


Graph 3A: Overall Acceptance Rate

Observation: Out of all applications, 40.3% are Rejected, 37.5% are Under Review, and only 22.2% are Selected.

Takeaway: The overall selection rate is low. The hiring process is competitive and most candidates are either rejected or still pending.

```
In [38]: # Per-domain - grouped bar chart
status_by_domain.plot(kind='bar', figsize=(9,5), color = ('#4ECDCA', '#FFE66D', '#1
plt.title('Application Status by Domain')
plt.xlabel('Domain')
plt.ylabel('Number of Applicants')
plt.xticks(rotation=45, ha='right')
plt.legend(title='Status')
plt.tight_layout()
plt.show()
```



Graph 3B: Acceptance Rate per Domain

Observation: UI/UX Design has the highest number of Selected candidates (5) and also highest Under Review (9). Data Science has the most Rejections (9).

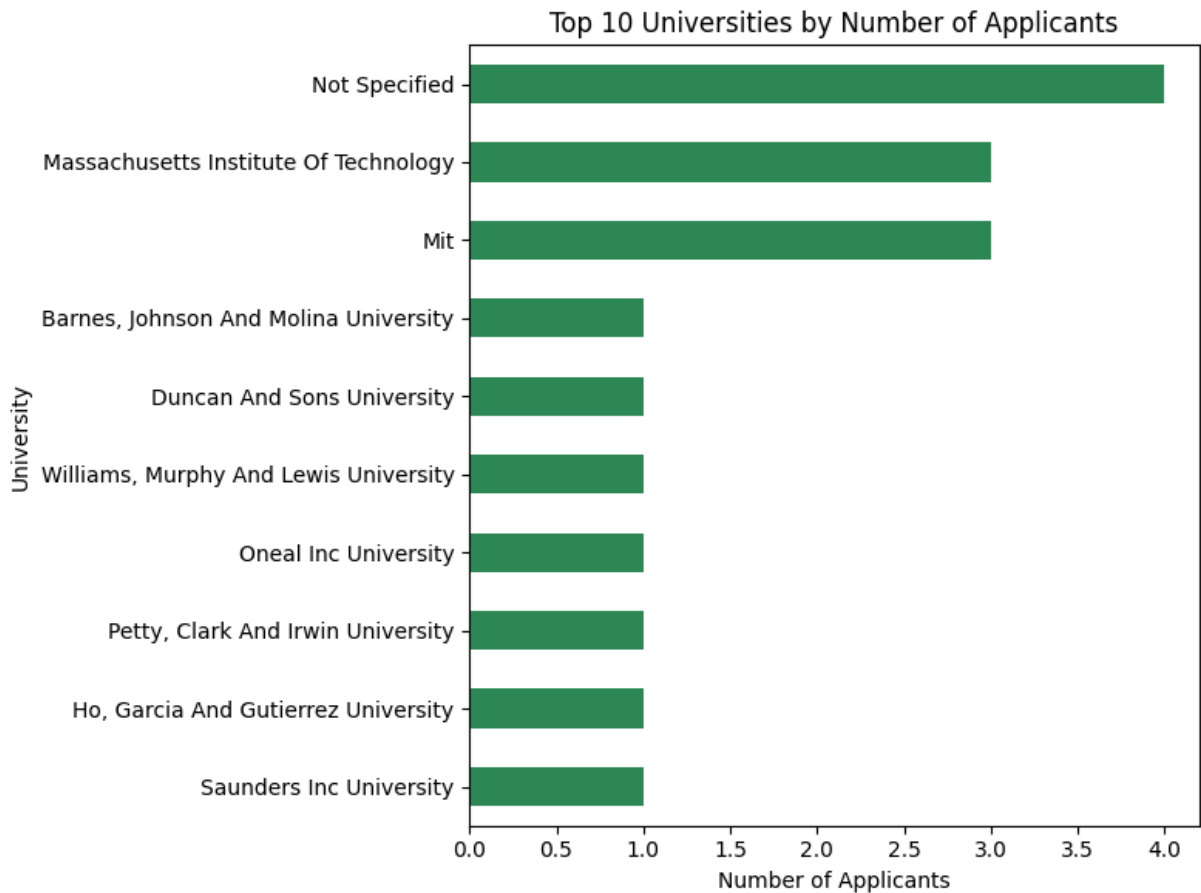
Takeaway: UI/UX Design is performing best in selections, while Data Science needs review of its rejection criteria or applicant quality.

Graph 4: Which Universities Sent the Most Applicants?

```
In [39]: top_universities = df['University'].value_counts().head(10)
print(top_universities)
```

```
University
Not Specified      4
Mit                3
Massachusetts Institute Of Technology  3
Saunders Inc University  1
Oneal Inc University  1
Petty, Clark And Irwin University  1
Ho, Garcia And Gutierrez University  1
Williams, Murphy And Lewis University  1
Duncan And Sons University  1
Barnes, Johnson And Molina University  1
Name: count, dtype: int64
```

```
In [40]: plt.figure(figsize=(8,6))
top_universities.sort_values().plot(kind='barh', color='seagreen')
plt.title('Top 10 Universities by Number of Applicants')
plt.xlabel('Number of Applicants')
plt.ylabel('University')
plt.tight_layout()
plt.show()
```



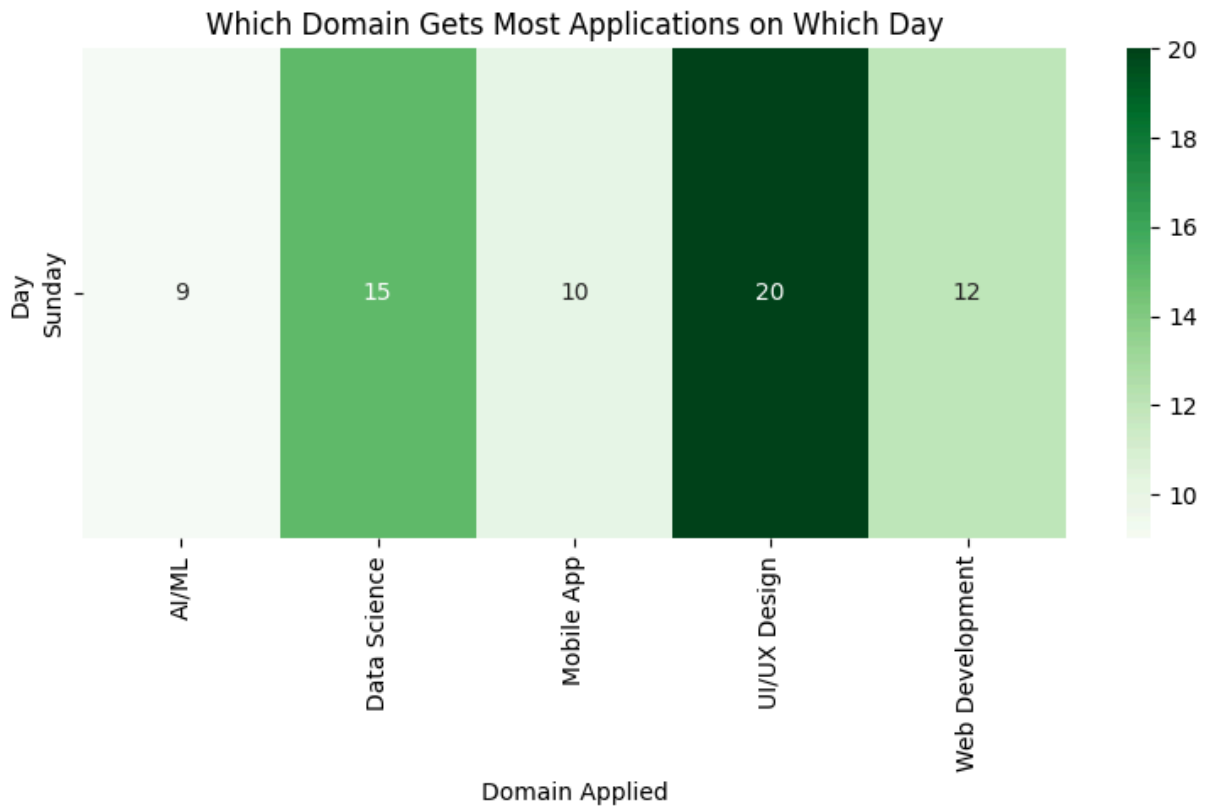
Observation: Most applicants did not specify their university. MIT have 3 applicants each.

Takeaway: Outreach should focus on collecting complete university data for better insights.

Bonus Chart: Domain Applications by Day of Week

```
In [41]: df['Day'] = df['Application Date'].dt.day_name()
heatmap_data = pd.crosstab(df['Day'], df['Domain Applied'])

plt.figure(figsize=(8,5))
sns.heatmap(heatmap_data, cmap='Greens', annot=True, fmt='d')
plt.title('Which Domain Gets Most Applications on Which Day')
plt.tight_layout()
plt.show()
```

Why I chose this:

I chose this heatmap to see which day gets the most applications for each domain. This helps in understanding student behavior and planning better campaign timings.

Observation:

From the heatmap, we can see that Sunday is the only day with application data. On Sunday, UI/UX Design received the highest 20 applications, followed by Data Science with 15. AI/ML had the lowest with 9 applications.

Takeaway:

All applications in the current dataset were submitted on Sunday, with UI/UX Design being the most popular domain on that day. This suggests weekends are critical for application submissions.

Key Insights

- Top Domain:** UI/UX Design is the most popular domain with 22 applications, followed by Data Science with 16. Mobile App has the lowest interest with only 10 applications.
- Lowest Domain:** Mobile App has the fewest applications with 10, followed by Web Development with 12, suggesting a need for more outreach in these areas.
- Selection Rate:** Only 22.2% of total applicants were Selected. 40.3% were Rejected and 37.5% are still Under Review, showing a competitive process.
- Domain-wise Acceptance:** UI/UX Design has the highest number of Selected candidates. Data Science has the highest rejection rate with 9 rejections out of 16

applications.

5. **Monthly Data:** All 66 applications were submitted in July 2026. Only 1 month of data is available, so trends cannot be analyzed yet.
6. **Peak Submission Day:** Sunday is the peak day for applications. UI/UX Design received 20 applications on Sunday, indicating weekend activity is highest.
7. **University Gap:** Most applicants did not specify their university. This is a data quality issue that needs to be fixed.
8. **Top University:** Among applicants who specified, MIT has the highest representation with 3 applicants, making it the top named university.

In []: